

Arduino I-V rig: the breadboard, hole for hole

This is the wiring in 2_ARDUINO_IV.md drawn out. Nothing is added and nothing is left out: 11 male-male jumpers, 2 female-male, 7 resistors, 3 caps, one Arduino UNO, one breadboard. Rail holes follow the column numbers: first one under column 3, then five in a row, a one-hole gap, five more, and so on.

THE BUILD

Arduino I-V rig - build this

Top view, drawn hole for hole. Only the bottom half of the breadboard is used.

- build it in this order
- 1

six bank resistors: row A col 14..24, other leg FORCE one col right
- 2

jumper D2..D7 into row E of its own bank column
- 3

link E4 to E8 - all ten of those holes are now RETURN
- 4

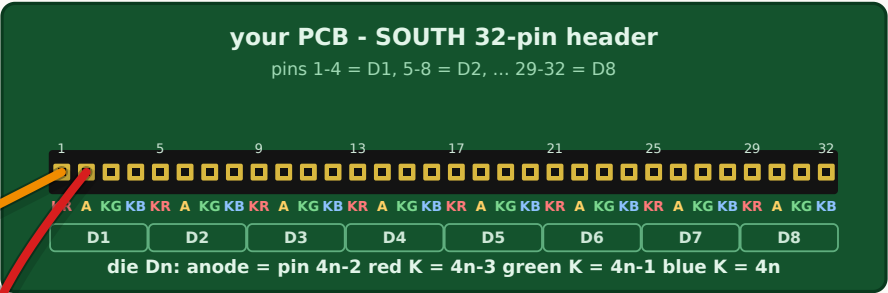
100 R from B8 down to the GND rail
- 5

C2 from A4, C3 from A8 to GND rail; C1 across the rails (68 nF)
- 6

jumper UNO GND to the GND rail col 3 - the only ground wire
- 7

jumper A0 to D4, A2 to D8, A1 to FORCE col 6, A3 to GND col 10
- 8

two F/M jumpers out to the die you are testing



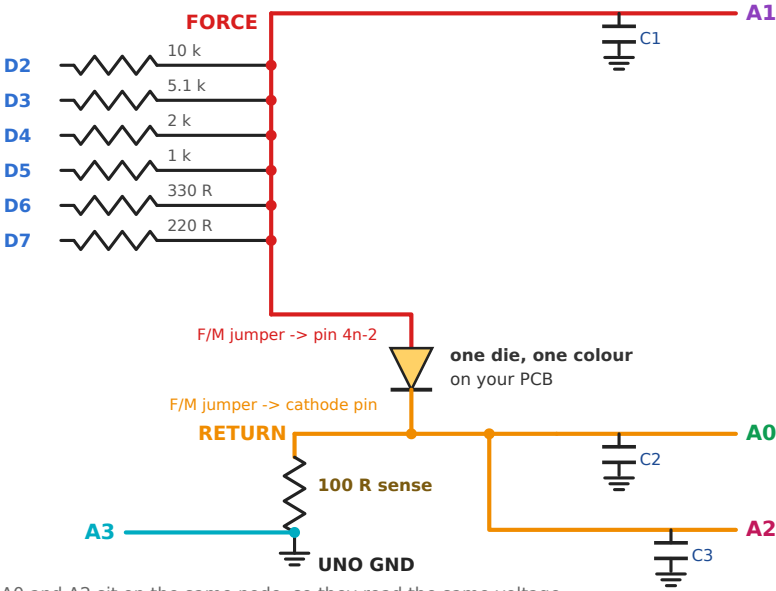
every wire, by colour

- 6 x UNO D2..D7 -> row E, cols 24 22 20 18 16 14
- 1 x UNO GND -> GND rail, col 3
- 1 x UNO A0 -> RETURN, hole D4
- 1 x UNO A1 -> FORCE rail, col 6
- 1 x UNO A2 -> RETURN, hole D8
- 1 x UNO A3 -> GND rail, col 10 NEW
- 1 x link E4 - E8 (RETURN becomes one node)
- F/M FORCE col 29 -> die anode, pin 4n-2
- F/M RETURN, C4 -> die cathode you are testing

check before you power up

- 12 male-male jumpers: 6 bank + A0 A1 A2 A3 + GND + 1 link
- 2 female-male jumpers - the only wires leaving the board
- 7 resistors: six in the bank, one 100 R RETURN to GND rail
- 3 caps, 68 nF each: C1 rail to rail, C2 + C3 on RETURN
- nothing in series between an ADC pin and the node it reads
- UNO GND touches the breadboard exactly once
- A0 and A2 land on the same node - that is intended
- A3 reads the GND rail: it cancels the ADC offset
- rail holes line up with the columns, starting at col 3
- each rail is one strip, so any hole on it is the same node
- the grey dash is only for boards with split rails
- To change channel: move only the two female-male jumpers.

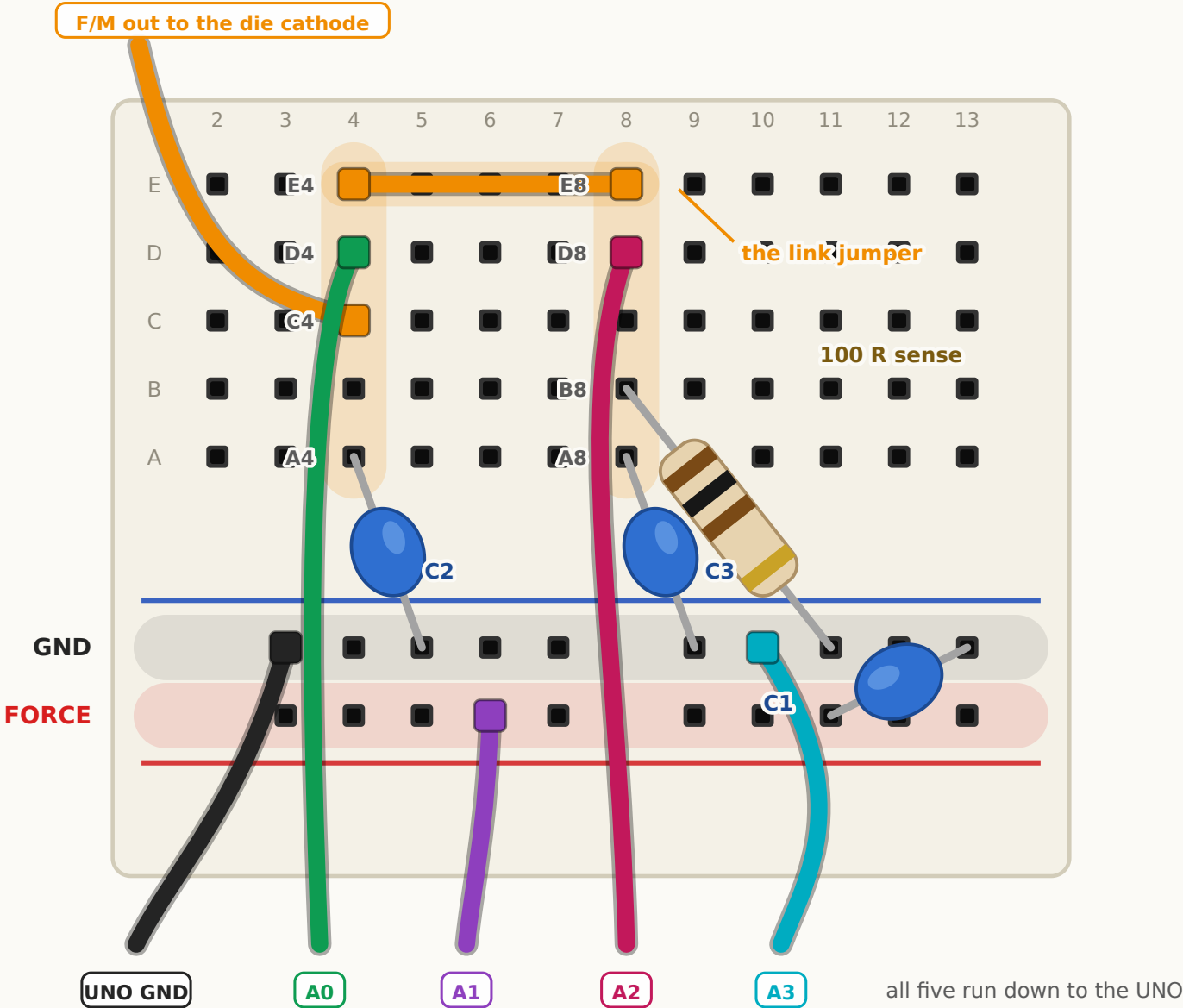
the same thing as a circuit



A0 and A2 sit on the same node, so they read the same voltage.
 $i = (A2 - A3) / R_SENSE$. The difference cancels the ADC offset.

Close-up: the RETURN cluster and the two rails

Columns 2 to 13 at twice the size. This is the part that was crowded.



The orange shading is one single node

Holes A-E of one column are already joined inside the board. The link jumper joins column 4 to column 8, so all ten of those holes are RETURN. Anything you plug into any of them is on the same node.

Three things there, three different jobs

C4 - female-male jumper, leaves the board for the cathode pin of the channel under test
E4 to E8 - the link, lies flat on the board
D8 - goes to UNO A2, drawn pink so it cannot be confused with the two orange ones

How the rail holes are laid out

First rail hole is under column 3, then five in a row, a one-hole gap, five more, and so on. They line up with the numbered columns. The whole strip is one node, so the exact hole never changes the circuit.

What ends up on each rail

A3 taps the GND rail in the SAME five-hole group as the 100 R, so $i = (A2 - A3) / R_SENSE$ is a true difference
GND rail - UNO GND, A3, C1, C2, C3, the 100 R
FORCE rail - UNO A1, C1, the six bank resistors, and the female-male jumper to the die anode

Columns 2 to 13 at twice the size. The orange shading is one node: five holes per column joined inside the board, and the link jumper ties column 4 to column 8. Note the rail gap at column 8, which the A2 wire runs straight through.

EVERY HOLE, BY NET

	FORCE		GND		RETURN
	bottom outer rail		bottom inner rail		cols 4 + 8, rows A-E
col 6	jumper to UNO A1	col 3	jumper to UNO GND — the only ground wire in the rig	A4	C2 down to the GND rail
col 11	C1, other leg on the GND rail at col 13	col 5	C2, other leg in A4	C4	F/M jumper out to the cathode of the channel under test
15 17 19	bank resistors, lower legs	col 9	C3, other leg in A8	D4	jumper to UNO A0
21 23 25	bank resistors, lower legs	col 11	100 Ω sense, other leg in B8	E4 ↔ E8	the link that makes both columns one node
col 29	F/M jumper out to the die anode , header pin 4n-2	col 10	jumper to UNO A3 — senses the bottom of the 100 Ω	A8	C3 down to the GND rail
7-9	a link across the gap, <i>only</i> if your board's rails are split rather than continuous	col 13	C1, other leg on the FORCE rail at col 11	B8	100 Ω sense down to the GND rail
	The rail is one strip, so any hole on it is electrically the same. These columns just keep the leads short.		Column 3 is the first hole the rail has. Columns 8, 14, 20 and 26 are the gaps and have no hole at all.	D8	jumper to UNO A2
					These eight hole numbers are the only ones that matter in the whole build.

THE RESISTOR BANK

Each branch is one jumper into row E and one resistor from row A across to the FORCE rail one column to the right. The diagonal keeps the resistor body clear of the GND rail so you can see at a glance that it is not plugged into it.

UNO PIN	JUMPER LANDS	RESISTOR	ROW A	LOWER LEG ON FORCE	BRANCH CURRENT, RED
D7	E14	220 Ω	A14	col 15	10.06 mA
D6	E16	330 Ω	A16	col 17	7.49 mA
D5	E18	1 kΩ	A18	col 19	2.93 mA
D4	E20	2 kΩ	A20	col 21	1.53 mA
D3	E22	5.1 kΩ	A22	col 23	0.62 mA
D2	E24	10 kΩ	A24	col 25	0.32 mA

COUNT BEFORE YOU POWER UP

12

male-male jumpers: 6 bank, A0, A1, A2, **A3**, GND, 1 link

2

female-male jumpers, the only wires leaving the board

7

resistors: six in the bank, one 100 Ω sense

3

capacitors, 68 nF each: C1 rail to rail, C2 and C3 on RETURN

- 1
- Six bank resistors: one leg in row A at columns 14, 16, 18, 20, 22, 24, the other leg in the FORCE rail one column to the right.
- 2
- Jumper D2–D7 into row E of each bank column: D7 to E14, up to D2 at E24.
- 3
- Link E4 to E8. All ten of those holes are now RETURN.
- 4
- 100 Ω from B8 down to the GND rail at column 11.
- 5
- C2 from A4 and C3 from A8 down to the GND rail; C1 from FORCE col 11 across to GND col 13.
- 6
- Jumper UNO GND to the GND rail at column 3. No second ground wire anywhere.
- 7
- Jumper A0 to D4, A2 to D8, A1 to the FORCE rail at column 6, and **A3 to the GND rail at column 10**.
- 8
- Two female-male jumpers out to the die: FORCE to pin 4n–2, RETURN C4 to the cathode pin of the colour you are sweeping.

To move to the next channel, move only those last two jumpers. Everything else stays put for the whole session.

WHY A3 IS THERE

Step 3 was run on a 100 Ω dummy and the rig **failed** it. A pure resistor must fit $v_{die} = i \cdot R$ through the origin. It fitted 99.32 Ω but with a **+13.8 mV intercept**, which is not something a resistor can do.

LEVEL	BRANCH	I PREDICTED	I MEASURED	DEFICIT ACROSS R_SENSE
1	10 kΩ	0.4943 mA	0.3634 mA	+12.9 mV
4	2 kΩ	2.3030 mA	2.1916 mA	+11.0 mV
16	330 Ω	9.0636 mA	8.9553 mA	+10.7 mV
32	220 Ω	11.3966 mA	11.2726 mA	+12.2 mV

Constant in millivolts across a 30× current range, so it is an additive ADC offset of about 10 mV, not a resistance or reference error, which would both scale with current. A diagnostic sweeping ADC prescaler (/32, /64, /128) against discarding a conversion after each mux change — six combinations — gave the same offset every time, so it is neither clock speed nor mux settling.

The offset cancels in v_{die} , because that is a difference of two channels. It does not cancel in the current, which was an absolute voltage. At the bottom of the sweep the current was wrong by **26 %**.

A3 makes the current a difference too. It taps the GND rail in the same five-hole group as the 100 Ω's lower leg, and the sketch computes $i = (v_{A2} - v_{A3}) / R_{SENSE}$. The offset subtracts out exactly, and the IR drop in the single GND jumper leaves the measurement with it.

ALSO FOUND: 2.9 Ω IN THE E4–E8 LINK

A0 and A2 are the same node by design and must read identically. A0 read high by an amount strictly proportional to current, $2.9 \pm 0.2 \text{ } \Omega$ across levels 2 to 32. That is contact resistance in the link jumper or its sockets. In this topology it sits outside both the v_{die} loop and the current tap, so it is not corrupting R_s , but it is thirty times the bond resistances being measured and a joint that loose will not stay put. Reseat it.

CAPACITOR VALUE

68 nF for all three. Anything from about 47 to 100 nF works and the tolerance does not matter, since nothing in the measurement depends on the value. Put the smallest of the three on C1, which is the only one whose settling is tight.

C1	$T = 10\text{ k}\Omega \cdot C$	SETTLE, RED $V_F \approx 2\text{ V}$	SETTLE, BLUE $V_F \approx 3\text{ V}$
68 nF	0.68 ms	0.35 ms	0.63 ms
100 nF	1.0 ms	0.51 ms	0.92 ms
220 nF	2.2 ms	1.1 ms	2.0 ms
1 μ F	10 ms	5.1 ms	9.2 ms

The slow node is FORCE with only the 10 kΩ branch on, at the bottom of the sweep, charging through $t = R \cdot C \cdot \ln(5/(5-V_F))$. The sketch allows `SETTLE_MS = 2`, so 100 nF is the ceiling. Below about 30 nF the ADC's 14 pF sample-and-hold starts stealing a visible fraction of the charge on every one of the 256 oversampled reads.

§4.5 OF THE BACKGROUND DOC SAYS UP TO 1 MF. IT CANNOT.

At 1 μF a blue channel needs 9.2 ms to reach V_F on the lowest current step, against a 2 ms settle. The ADC would read a node still on its way up, the fit would see too little voltage at low current, and that lands squarely on R_s as series resistance that is not there. The same section's own worked example (0.7 ms at 68 nF, comfortably inside 2 ms) is the reasoning that rules out its own upper bound.

TWO THINGS IN THE DOC THAT DO NOT AGREE

THE BUILD BLOCK STILL LISTS THE OLD RESISTOR VALUES

The parts table in `2_ARDUINO_IV.md` says 10 k, 5.1 k, 2 k, 1 k, 330 Ω, 220 Ω, and the current table underneath it (0.32 / 0.62 / 1.53 / 2.93 / 7.49 / 10.06 mA) is computed from exactly those. The ASCII Build block a few lines lower still says 4k7, 2k2 and 470, carried over from `measurements/ARDUINO_IV_RIG.md`. The drawing uses the parts table. The ASCII block is the one that needs editing.

A0 AND A2 LAND ON THE SAME NODE, SO THIS IS NOT KELVIN SENSING

Section 4.2 of `ARDUINO_IV_RIG.md` puts A1 and A0 on the Tier-1 probe pads so that jumper wire and header contact resistance sit outside the sense loop. Round 2 instead takes both from the breadboard, which folds both female-male jumpers and both header contacts back into $V(A1) - V(A0)$. That is a few tens to a couple of hundred mΩ, the same size as the bonds you are chasing.

It is not fatal, and it is the reason the plan already asks for seat2 and seat3 repeats: the wiring term is what re-seating changes. But quote R_s from this rig as die + wiring, and treat the seat-to-seat spread as the error bar on the wiring part rather than as noise.

Drawn from FINAL_MEASUREMENTS/2_ARDUINO_IV.md · pad mapping from round 1: pad 1 = red cathode, pad 2 = anode, pad 3 = green, pad 4 = blue · the “A KG KB KR” order silkscreened on board-probe-map.png is the old, wrong one.